Leveraging the mathematical framework of Geshkovski et al. (2023), in the continuous approximation of a forward pass through deep residual network layers as evolution through time (Chen et al., 2018), the dynamics of the representations for token i, z_i can be modeled as

$$\dot{z}_i(t) = \mathbf{P}_{z_i(t)} \left[\frac{1}{Z_{\beta,i}(t)} \sum_{j=1}^{n} \exp \left\{ \beta \left[W_Q(t) z_i(t) \right]^T W_K(t) z_j(t) \right\} W_V(t) z_j(t) \right], \tag{14}$$

where, $Z_{\beta,i}(t)$ is the partition function, β the system's *temperature*, and for d-dimensional z and u, $\mathbf{P}_z u = u - z^T u z$ is the projection of u onto the tangent space of a $(d-1)$-dimensional sphere in R^d at z (Geshkovski et al., 2023). The similarity between particle systems and attention mechanisms is evident from comparing equations 1 and 14. Thus, the output of the simplified transformer studied here can be approximated as the solution of Equation 14 as $t \to +\infty$. This framework sets the stage for our empirical investigation (Section 5) into how ORC distribution informs the behavior of transformers and their robustness properties in light of the discussion in Section 2.

Forward pass and attention clustering. Geshkovski et al. (2023) demonstrate that when the query, key, and value projection matrices are the identity, the dynamics modeled by this equation are similar to the Krause model (Krause et al., 2000), which shows particles clustering when $t \to +\infty$ (Jabin & Motsch, 2014). Clustering behavior is also observed in other similar models (Geshkovski et al., 2023; Motsch & Tadmor, 2014). Despite the differences between transformer models used in practice and the idealized model commonly assumed for analysis, the interpretation above leads to questions on attention clustering in transformers. Geshkovski et al. (2023) show empirical evidence that in commonly used pre-trained transformers the attention map becomes more clustered through a forward pass. Given the connection between ORC and clustering in networks (Jost & Liu, 2014; Sia et al., 2019), which we expand upon in Appendix C, the forward pass dynamics of attention also corresponds to an evolving pattern of ORC distribution.

Attention dynamics and Ricci curvature. Computing the ORC distribution in pre-trained and fine-tuned vision and language transformers, our experiments confirm our theoretically-anticipated connection between attention ORC distributions and forward pass dynamics in transformers. Empirical observations reveal that a forward pass in transformers leads to a shift in the ORC distribution towards more positive values, or significantly reduces the frequency of smaller ORC values. This is visualized for two vision and two language transformers with six standard datasets in Figure 8, and quantified by the Wasserstein distance of the distribution from a Dirac mass at 1 – maximum ORC value. A smaller distance indicates a more positive-leaning distribution in the final attention layer compared to the first layer, as observed in most cases. In the only exceptions where this Wasserstein distance value is greater – DeiT model on CIFAR10 and BERT-Tiny on IMDb – we still observe that the forward pass eliminates the smaller peak in the ORC distribution, shifting the smaller values upwards. As we explain in Section 2, ORC is a known indicator of system robustness. Meanwhile, the forward pass in transformers is known to play an important role in their remarkable performance, for instance, in their in-context learning (Von Oswald et al., 2023). Hence, these evolving patterns of ORC which follow transformers' dynamics reveal evidence on the geometric signature of both the robustness and performance of transformers.

C Graph Ricci curvature and weight distributions

The Ollivier-Ricci curvature for any given edge in the graph depends on the geodesic distances of nodes in the neighborhood of the edge. This is evident from the definition of ORC in Equation 2. To better explain this point, consider an edge $(v, u) \in \mathcal{E}$ in a graph $G = (\mathcal{V}, \mathcal{E})$, and let us expand the Wasserstein distance in Equation 2 to write the ORC in the form

$$\kappa_{OR}(v, u) \coloneqq 1 - \frac{1}{d_G(v, u)} \inf_{\gamma \in \Gamma(\mu_v, \mu_u)} \int_{\mathcal{N}_v \times \mathcal{N}_u} d_G(v', u') \, d\gamma(v', u'), \tag{15}$$

where $d_G(\cdot, \cdot)$ is the geodesic distance over G, $\mathcal{N}_v$ and $\mathcal{N}_u$ denote the neighborhoods of nodes v and u, respectively, μ_v and μ_u are probability measures over $\mathcal{N}_v$ and $\mathcal{N}_u$ and $\Gamma(\mu_v, \mu_u)$ is the set of all couplings over them.

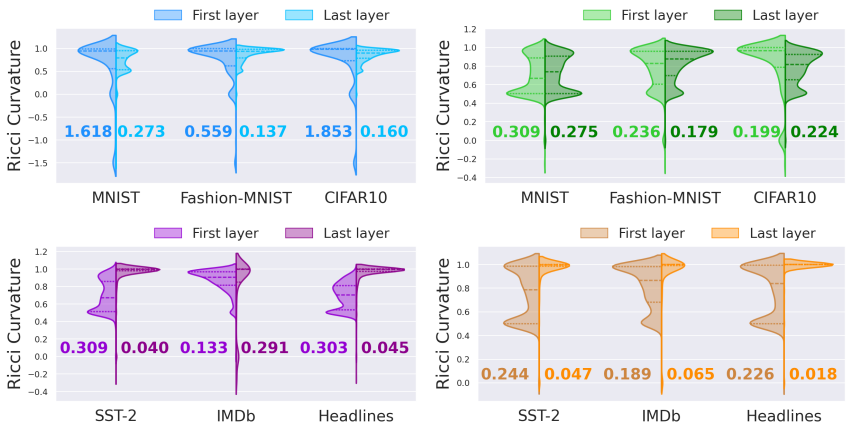

Figure 8: The ORC distributions of the first and last attention layers of ViT (blue) and DeiT (green) for a batch of MNIST, Fashion-MNIST, and CIFAR10 images, and of BERT-Tiny (purple) and ELECTRA-Small (orange) for a batch of SST-2, IMDb, and Headlines data. The curvature distribution shifts towards more positive values through a forward pass; quantified by the Wasserstein distances of the ORC distributions from a Dirac mass at the ORC value of 1, annotated next to each distribution (smaller means more positive-leaning).

Curvature and clustering. Note that for any pair of nodes w and w', $d_G(w, w')$ is the weighted shortest path between w and w'. Hence, given two edges (v, u) and (v', u'), the similarity of weight distributions over the neighborhoods of (v, u) and (v', u') leads to the similarity of geodesics in their neighborhoods, and considering the definition in Equation 15, this results in the proximity of the values of $\kappa_{OR}(v, u)$ and $\kappa_{OR}(v', u')$. Geshkovski et al. (2023) find that in a forward pass, attention maps exhibit clustering signaled by the cosine similarity of pairs of nodes, which corresponds to the similarity of weight distributions in their neighborhood. Given the discussion above, this in turn leads to a change in the ORC pattern, as a group of edges with similar ORC values form. This indeed matches the observation in Figure 3, showing the emergence of more concentrated modes in the distribution of ORC following attention dynamics through a forward pass.

Curvature and weight variance. Recall that the ORC definition provided in Equation 15 expands the Wasserstein distance from Equation 2, which is the cost of optimal transport of a mass distribution from $\mathcal{N}_v$ to $\mathcal{N}_u$. Recognizing this, we can observe that, fixing any (unweighted) connectivity structure, the higher the imbalance in the neighborhoods, the larger their distributional distances, which leads to smaller curvature values considering the definition of ORC. Note that the connectivity in full attention is trivially fixed, since the graph is always fully connected. It follows that the curvature, in this case, depends on the (im)balance in weight distributions. Therefore, it is the *variance* of weight distribution that determines the curvature values in attention maps. This intuition informs the regularization proposed in Section 4.2 for training curvature-adjusted transformer models.

D Implementation Details

D.1 *L-ViT* Configuration

The *L-ViT* model utilizes a transformer architecture for image classification borrowed from ViT-PyTorch library (Wang, 2020). The input image is divided into 7x7 patches for the MNIST and Fashion-MNIST datasets, and 8x8 patches for the CIFAR10 dataset. These are projected into a 64-dimensional space within a single channel for the MNIST and Fashion-MNIST datasets, and 3 channels for the CIFAR10 dataset. An 8-layer deep transformer encoder with a single attention head per layer processes these patch embeddings. The MLP block within each layer has a hidden dimension of 128. Dropout with a rate of 0.1 is applied both to activation functions and embeddings during training to enhance model robustness and prevent overfitting.

D.2 Training and evaluation details

We use a learning rate of $5 \cdot 10^{-4}$ for training *L-ViT* with a batch-size of 64 images, and a learning rate of $5 \cdot 10^{-6}$ for training BERT-Tiny with a batch-size of 16 sentences. The pre-trained model weights were downloaded from Huggingface Transformers (Wolf et al., 2020). For fine-tuning pre-trained vision models, we use a learning rate of $2.5 \cdot 10^{-4}$ with a batch-size of 64 images, and for pre-trained language models, these values are $2 \cdot 10^{-5}$ and 16.

The perturbations for experiments in Section 5 include flipping, blurring, and rotation of images with probabilities 0.2 and 0.4, and word swap in texts with probability 0.4. For the experiments in Section 4.2, we trained *L-ViT* for 20 epochs on MNIST and Fashion-MNIST images and 40 epochs on CIFAR10 images, and similar experiments on BERT-Tiny (in Appendix E.1) were performed with 20 epochs for all three datasets. For the *L-ViT* experiments in Appendix E.3, models were trained with an early stopping condition at accuracy of 97%, 87%, and 50%, over MNIST, Fashion-MNIST, and CIFAR10 validation sets, respectively. BERT-Tiny models mentioned in Appendix E.3 were trained with an early stopping condition at accuracy of 88%, 84%, and 79%, on the IMDb, Headlines, and SST-2 validation sets, respectively. In Appendix E.3, we perturb the test set images by applying elastic transformation with $\alpha = 75$ and random perspective transformation with a distortion scale of 0.5, with a probability of $p = 0.5$. Text test set sentences are perturbed via characters or word swap. These substitutions were chosen randomly, potentially resulting in nonsensical combinations. This process aimed to introduce noise and potentially disrupt the inherent coherence and structure of the sentences. Both character-level and word-level perturbations were applied independently with a fixed probability of $r = 0.2$ for each. Consistent subset sizes were employed for all experiments involving ORC computation over subsets. For both image and text datasets evaluated with *L-ViT* and BERT-Tiny, we used a fixed subset of 1,024 samples.

D.3 Other implementation details and system specification

All ORC computations in this study were conducted using the GraphRicciCurvature(Ni et al., 2019) Python library. All loss optimizations were performed via Adam (Kingma & Ba, 2014), and the training and testing implementations use the PyTorch library (Paszke et al., 2019), with the default data split from the Torchvision package (Marcel & Rodriguez, 2010). While visualizing distributions, outliers and zeros were removed as needed for clarity of the distributions in the figures. The system specification for the computations is provided in Table 2.

Table 2: System specifications for the computations.

CPU	Intel(R) Xeon(R) CPU @ 2.40GHz
GPU	Nvidia A100 SXM4 40GB
OS	AlmaLinux 9.3
Architecture	x86_64

Due to computational costs associated with the resource demands of graph ORC computations, evaluations of several experiments were conducted on randomly selected subsets of the MNIST, Fashion-MNIST, and CIFAR10 image datasets and the IMDb, Headlines, and SST-2 text datasets. Additionally, the pre-trained models were fine-tuned for up to 2 epochs on each dataset, which already yields over 90% accuracy.

E Additional experiments

E.1 Curvature-adjusted training of language transformers

In Section 4.2 we propose a regularization method to adjust the curvature distribution of the attention map in transformers. This adjustment is motivated by our theoretical results in Section 4.1, and the proposed variance regularization follows the intuition described in Appendix C. The experiments in Section 4.2 further demonstrate the impact of this regularization, which, in addition to validating our theoretical results, provides a tool for training better-performing or more robust transformers. In this appendix section, we repeat these experiments for a language model.